

High Bandwidth Control of Ball Screw Drives

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Abstract

This paper presents a tracking control strategy for high speed ball screw drives. Rigid body motion is controlled using adaptive sliding mode control. Torsional vibrations are modeled, experimentally identified, and compensated in the control law using notch filtering and active cancellation techniques. Attenuation of torsional resonances improves the stability margins and enables high positioning bandwidth to be achieved. The axis friction is experimentally identified and cancelled out in feedforward, in order to improve the positioning accuracy at motion reversals. The developed control law is verified experimentally on a high speed ball screw drive, where a tracking accuracy of 1.6 μm is maintained while traversing the axis at 1000 mm/sec feed and 1 g acceleration.

Keywords:

Feed, Drive, Control

1 INTRODUCTION

Ball screw drives are among the most commonly used motion delivery mechanisms in high speed machine tools. Their high stiffness allows accurate tool positioning in the presence of cutting forces. Their high speed positioning accuracy is determined by the bandwidth of the servo control loop [1], which is usually limited by structural resonances [2], [3], [4]. The aim of this paper is to achieve high accuracy motion through high bandwidth control of ball screw drives. This is realized by mitigating the adverse effects of structural vibrations, and designing a robust sliding mode controller which can deal with parameter variations and external disturbances.

Torsional and axial vibrations of the ball screw cause oscillatory behavior when the bandwidth of the control loop is increased [4]. Feedforward prefiltering of trajectory commands [5], [6], [7], [8], or notch filtering of the control signal [9] helps to partially alleviate this problem, by reducing the excitation delivered through the control signal. However, this attenuation is not perfect, as structural frequencies can also be excited by external disturbances such as cutting forces or sudden friction transients. In addition, the location and damping of structural resonances may vary as a function of the machine's configuration [10]. Hence, the control law needs to possess robustness against varying or unmodeled axis dynamics [11], [12], as well as the ability to attenuate structural vibrations [13], [14] and counteract external disturbances, such as cutting forces [15] and friction [16].

The axis friction can be modeled and compensated in feedforward to a certain extent [17], which helps mitigate the quadrant errors in contouring operations. The unpredictable disturbance needs to be estimated and cancelled out in real time, which has been successfully implemented in various disturbance adaptation strategies [18], [19], [20]. Applying disturbance adaptation not only improves the tracking accuracy, but also provides robustness against drive parameters variations, that are perceived as equivalent disturbances within the adaptation bandwidth, and cancelled out. This results in the closed loop system to respond in accordance with its nominal design, in spite of variations in the open loop dynamics.

In this paper, a high bandwidth control scheme is proposed for ball screw drives. Dynamics of the drive system, comprising of rigid body motion, friction, and torsional vibrations, are modeled and identified in Section 2. The controller design, which consists of using a combination of active and passive damping techniques to attenuate the structural vibrations, and adaptive sliding mode control [20] to handle the rigid body motion, is presented in Section 3. Effectiveness of the proposed control strategy is verified in high speed tracking tests and frequency response experiments in Section 4. The conclusions are presented in Section 5.

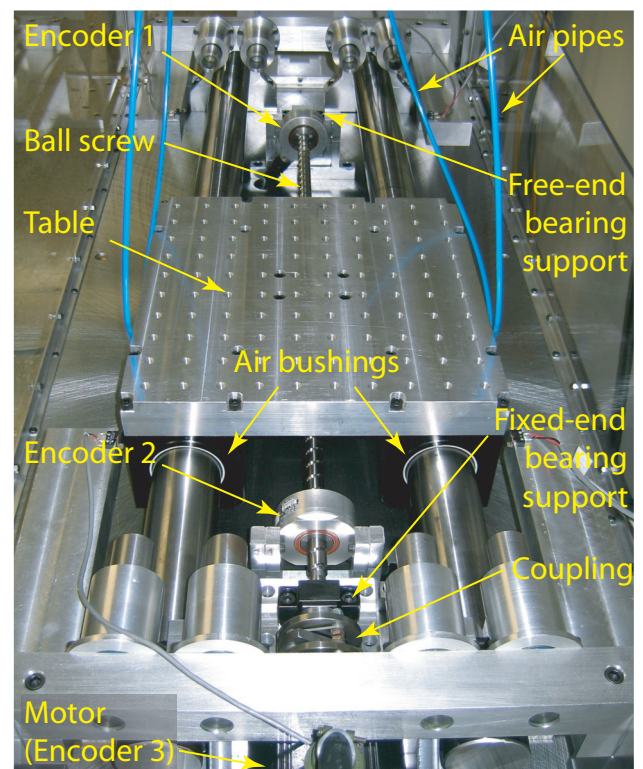


Figure 1: Ball screw drive setup.

2 EXPERIMENTAL SETUP

The high speed ball screw drive built at the Precision Controls Laboratory, University of Waterloo is shown in Figure 1. The ball screw, which has 20 mm diameter and 20 mm pitch, is used to drive a table supported on an air guideway system. Power is transferred to the mechanism through a 3 kW AC servomotor and a non-backlash coupling. The motor is operated in current control mode and the bandwidth of the current control loop was measured to be 440 Hz.

The ball screw is instrumented at both ends with two high resolution rotary encoders which deliver 5000 sinusoidal waves per revolution. These encoder signals are interpolated 400 times in the motion controller, resulting in a position measurement resolution equivalent to 10 nm of table motion. For successful vibration measurement, these encoders have been mounted rigidly onto the ball screw and to the machine base. Encoder 1 is located at the free end of the ball screw, and Encoder 2 is right after the fixed bearing support, as seen in Figure 1. A third encoder (Encoder 3) is available on the back of the motor which delivers 16384 quadrature pulses, resulting in a position measurement resolution equivalent to 305 nm of table motion.

The high resolution of the encoders enables accurate detection of the frequency response at different points along the ball screw, by yielding clear acceleration signals after double differentiation. Hence, the torsional mode shapes can be observed and verified up to a frequency of 2.5 kHz. The setup is controlled with a dSPACE controller where the identification and control algorithms have been implemented. The sampling frequency used is 20 kHz.

The rigid body dynamics and friction have been identified by applying a combination of least squares parameter estimation and Kalman filtering [17]. The equivalent inertia and viscous damping with respect to the motor shaft are identified as $J = 2.10 \times 10^{-3} \text{ kgm}^2$ and $B = 1.10 \times 10^{-3} \text{ kgm}^2/\text{sec}$ in the mentioned order. The amplifier gain and motor torque constant are rated as $K_a = 1.7193 \text{ A/V}$ and $K_t = 0.57 \text{ Nm/A}$ respectively. The friction disturbance torque, which originates mainly from the motor, support bearings, nut, and encoders, was identified by jogging the axis back and forth under control at various speeds and observing the control signal equivalent disturbance through a Kalman filter. The friction model was fit on the observations in the form of:

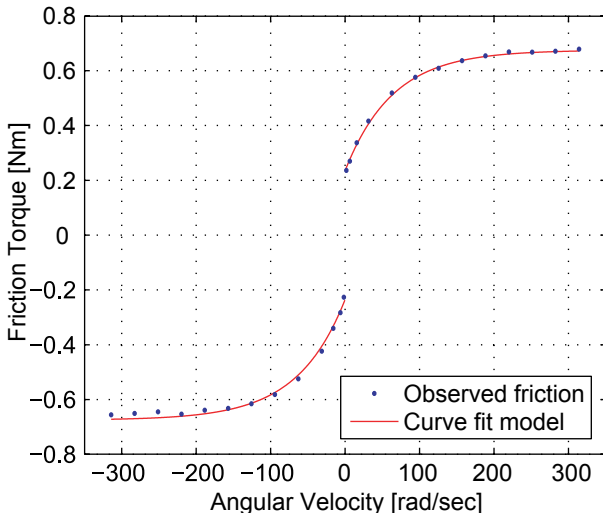


Figure 2: Experimentally identified friction model.

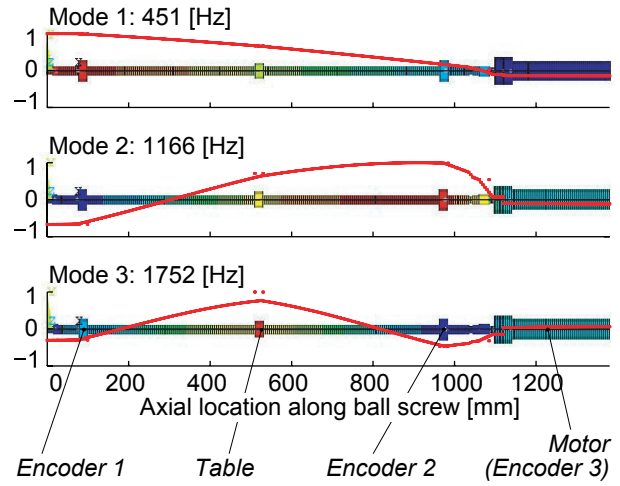


Figure 3: Torsional mode shapes predicted with FEA.

$$T_{fric} = T_s + T_d(1 - e^{-\omega/V_c}) \quad (1)$$

Above, T_{fric} Nm is the equivalent friction torque, ω rad/sec is motor angular velocity, $V_c = 64 \text{ rad/sec}$ is the velocity constant, $T_s = 0.235 \text{ Nm}$ is the static friction torque, and $T_d = 0.440 \text{ Nm}$ such that $T_s + T_d = 0.675 \text{ Nm}$ equals the Coulomb friction torque. The observed and curve-fit friction models are shown in Figure 2.

A Finite Element (FE) model has been constructed to investigate the torsional dynamics of the ball screw mechanism. The model includes the inertial characteristics of the motor, coupling, ball screw, encoders, bearing rings, table and nut assembly. The coupling and nut stiffness were substituted from rated catalogue values. The bearings were modeled by imposing appropriate boundary conditions. Due to the complexity of the motion delivery between the ball screw and the nut [21], which exhibits a hysteresis type of behavior, the equivalent inertia reflected from the linear motion of the table and nut was determined experimentally, to yield consistent modal frequencies with measured Frequency Response Functions (FRF's). All other parameters were determined analytically or substituted from nominal values.

The predicted mode shapes and natural frequencies are shown in Figure 3. In the 1st torsional mode (451 Hz), the neutral position is near the motor end of the ball screw. The torsional displacement gradually increases towards the free end where Encoder 1 is mounted, at which the maximum deflection occurs. In the 2nd torsional mode (1166 Hz), there are two neutral positions: one near the motor and the other between the table and Encoder 1. The maximum amplitude occurs near the location of Encoder 2. In the 3rd torsional mode (1752 Hz), three neutral positions occur, two between the table and Encoders 1 and 2, and the third near the motor. A positive peak occurs at the table location, and negative peaks occur close to the locations of Encoders 1 and 2.

The torsional vibrations were verified in frequency response tests, where a sinusoidal testing signal was applied to the amplifier input and rotational displacements were measured from Encoders 1, 2, and 3. The displacement measurements were double differentiated to obtain acceleration profiles, which were used in magnitude and phase evaluation with respect to the excitation signal. The acceleration FRF's measured from the three encoders are shown in Figure 4. These measurements

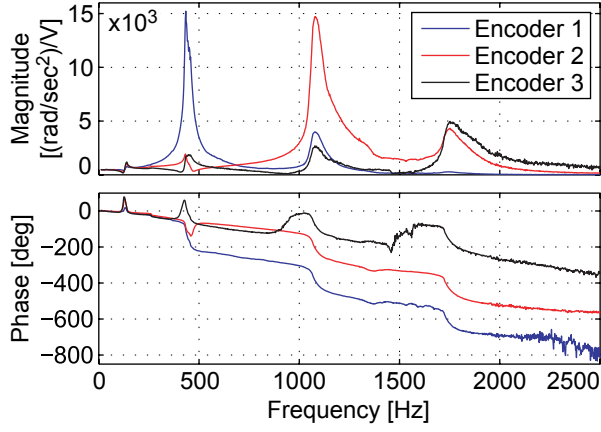


Figure 4: Measured open loop acceleration FRF's.

contain the dynamics of the ball screw mechanism, as well as the amplifier and the motor.

The first torsional resonance occurs at 445 Hz and is mainly observed by Encoder 1, where vibration amplitude is maximum. The amplitudes observed at Encoders 2 and 3 are smaller, which is in agreement with the 1st mode shape shown in Figure 3. The second resonance occurs at 1080 Hz which is observed by all encoders. Encoder 2 is subject to the highest magnitude of vibration, which can be explained with the 2nd mode shape. The third resonance is observed at 1755 Hz. The additional resonance seen at 140 Hz is attributed to the 1st mode of axial vibration of the ball screw and table assembly, which was analytically computed to be at 169 Hz. Overall, it can be said that the torsional mode shapes obtained from FE analysis are in close agreement with the experimentally measured FRF's.

Identification tests were also repeated with different excitation amplitudes, ranging from 2 to 8 Nm, and at different axial locations of the table. It was observed that the torsional FRF's were more or less consistent in spite of changes in the excitation amplitude and axis position, indicating reasonably linear behavior of the ball screw drive setup.

3 HIGH BANDWIDTH CONTROLLER DESIGN

The proposed controller design is shown in Figure 5. The objective is to achieve high positioning and disturbance rejection bandwidth, as well as robustness to parameter variations and structural vibrations. Adaptive Sliding Mode Control (SMC) is used in generating the primary control signal, which handles the rigid body motion. Sudden friction transients, which cannot be immediately detected by SMC's disturbance adaptation, are anticipated and cancelled out using knowledge of the reference trajectory and friction dynamics. High control bandwidth is achieved by attenuating the structural vibrations using a combination of notch filtering and active damping techniques. The implementation of each component is briefly explained in the following.

3.1 Adaptive Sliding Mode Control (SMC)

The idea of using sliding surfaces in controller design was pioneered by Utkin [22], which originally led to discontinuous control laws that were not practical for implementation on real machinery. Later, a general adaptive framework for SMC was developed by Slotine and Li [23], where a continuous and nonlinear control law is derived with Lyapunov stability, for adapting to parameters within known bounds [24]. When the adaptation is conducted only for the unknown disturbance,

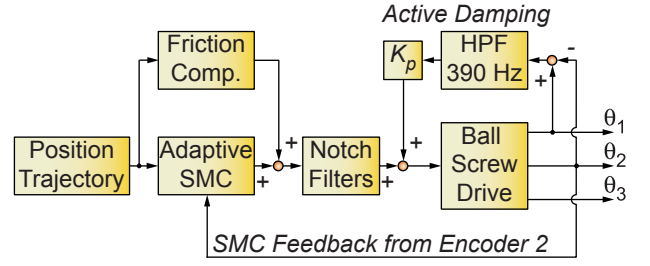


Figure 5: Proposed control strategy.

which is the case in this paper, the sliding mode control law assumes the below linear structure, as derived in [20]:

$$u_{smc} = K_s \sigma + J_e [\ddot{\theta}_{ref} + \lambda(\dot{\theta}_{ref} - \dot{\theta})] + B_e \dot{\theta} + \hat{d} \quad (2)$$

Above, θ_{ref} is the commanded and θ is the measured angular position of the ball screw. $J_e = J/(K_a K_t)$ and $B_e = B/(K_a K_t)$ are the control signal equivalent inertia and damping, which can be calculated from identified axis parameters. λ is the sliding surface bandwidth, σ is the deviation of position and velocity state errors from the sliding surface such that:

$$\sigma = \dot{\theta}_{ref} - \dot{\theta} + \lambda(\theta_{ref} - \theta) \quad (3)$$

$K_s > 0$ is a feedback gain to ensure Lyapunov stability, and $\hat{d}$ is the estimated equivalent disturbance which is used as the cancellation signal. The disturbance estimation is given as:

$$\dot{\hat{d}} = \rho \kappa \sigma \quad (4)$$

where $\rho > 0$ is the adaptation gain and κ is a binary variable to switch the adaptation on and off, in order to limit the disturbance adaptation within specified bounds of $\hat{d} \in [d^-, d^+]$:

$$\kappa = \begin{cases} 0 & \text{if } \hat{d}(k) \leq d^- \text{ and } \sigma \leq 0 \\ 0 & \text{if } \hat{d}(k) \geq d^+ \text{ and } \sigma \geq 0 \\ 1 & \text{otherwise} \end{cases} \quad (5)$$

In closing the position feedback loop, Encoder 2 is used which provides a direct measurement of the ball screw position, as opposed to using Encoder 3 on the motor, where the measurement also contains dynamics of the coupling and motor-encoder assembly. In addition, considering the open loop FRF's shown in Figure 4, the phase shift and resonance amplitude of the first torsional mode (445 Hz) registered through Encoder 2 are significantly lower compared to ones measured through Encoder 1, which enables higher control bandwidth to be achieved before encountering stability problems due to structural vibrations.

The SMC was tuned to yield smooth axis motion with minimal tracking error, which resulted in a sliding surface bandwidth of $\lambda = 1400$ rad/sec (223 Hz) to be achieved. This bandwidth has been experimentally verified in the closed loop tracking FRF, presented in Figure 9. The high control bandwidth could be achieved after implementation of the proper notch filters and vibration cancellation technique, presented in Sections 3.3 and 3.4. Without the vibration attenuation provided by these components, the sliding surface bandwidth could only be increased up to $\lambda = 450$ rad/sec (72 Hz).

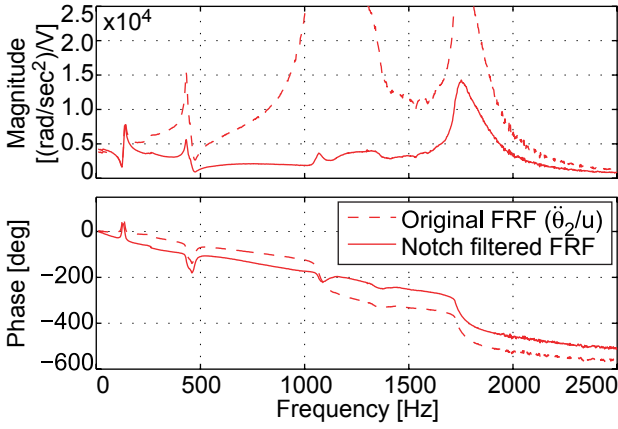


Figure 6: Original and notch filtered acceleration FRF (measured from Encoder 2).

3.2 Friction Compensation

The friction compensation signal is generated by substituting the commanded velocity $\dot{\theta}_{ref}$ into the friction model in Equation 1 and applying the necessary scaling to account for the motor torque and amplifier current constants:

$$u_{fric} = \frac{1}{K_a K_t} \left[T_s + T_d (1 - e^{-\dot{\theta}_{ref}/V_c}) \right] \quad (6)$$

3.3 Notch Filtering

The notch filters were designed to deliver a flat frequency response, by dampening out the first and second torsional modes, when cascaded with the axis dynamics observed from Encoder 2. Their combined transfer function can be expressed in the form:

$$N(s) = \prod_{k=1}^2 \left(\frac{\omega_{n,pk}^2}{\omega_{n,zk}^2} \cdot \frac{s^2 + 2\zeta_{zk}\omega_{n,zk}s + \omega_{n,zk}^2}{s^2 + 2\zeta_{pk}\omega_{n,pk}s + \omega_{n,pk}^2} \right) \quad (7)$$

The tuned filter frequencies and damping ratios are: $\omega_{n,z1}=432$ Hz, $\zeta_{z1}=0.45$, $\omega_{n,p1}=441$ Hz, $\zeta_{p1}=1.0$, $\omega_{n,z2}=1021$ Hz, $\zeta_{z2}=0.03$, $\omega_{n,p2}=1096$ Hz, $\zeta_{p2}=1.0$. The acceleration frequency response functions without and with notch filtering are shown in Figure 6.

3.4 Active Damping

The notch filters help avoid the excitation of the first two torsional modes by limiting the frequency content of the control signal. However, they provide no guarantee that these modes will not be excited by external disturbances, in particular sudden transients in the friction or cutting forces. Furthermore, considering the location of Encoder 2 on the first torsional mode shape (Figure 3), when the ball screw vibrates in this mode, Encoder 2 senses only a small portion of this vibration. Hence, the overall motion may appear to be smooth in the feedback loop, while the ball screw could be encountering structural vibrations at its first torsional mode. This case was frequently observed in experiments. To alleviate this problem, a vibration cancellation technique is incorporated into the control scheme.

The twist on the ball screw is computed by subtracting the measurement of Encoder 2 from Encoder 1. The torsional vibrations are decoupled from the steady state twist by applying High Pass Filtering (HPF) at $\omega_c=390$ Hz, which is just below the first torsional frequency. The cancellation

signal is generated proportionally to the filtered position difference between the two encoders, in order to counteract the vibration registered at the free end of the ball screw (Encoder 1) by replicating the same motion at the actuated end (Encoder 2). The cancellation signal is obtained as,

$$u_{damp} = K_p \frac{s^2}{(s + \omega_c)^2} [\theta_1 - \theta_2] \quad (8)$$

This approach was found to be effective in damping out the first torsional mode. The amplifier gain drops below -3 dB after 440 Hz, and the phase shift for the open loop FRF measured from Encoder 2, which is the controlled end, exceeds 180 deg after the second mode (Figure 4), making it difficult to dampen out the second and higher order modes. The vibration cancellation signal is injected after the notch filters, in order to avoid attenuation by these filters, as shown in Figure 5.

The overall control signal is obtained by combining the contribution of each component:

$$u = N(s) \cdot [u_{smc} + u_{fric}] + u_{damp} \quad (9)$$

4 EXPERIMENTAL RESULTS

The performance of the designed controller was verified in high speed tracking tests. A jerk continuous feed motion was commanded with 350 mm displacement, 1000 mm/sec velocity, 1 g acceleration, and 200,000 mm/sec³ jerk. The commanded trajectory, which delivers minimal high frequency harmonics due to its smooth nature, is shown in Figure 7.

The experimentally recorded tracking error and control signal profiles are shown in Figure 8. The position unit has been changed to μm , using the ball screw pitch length (20 mm/rev), in order to interpret the results in terms of contribution to the linear positioning accuracy. As can be seen from Figure 8, the maximum servo error does not exceed 1.6 μm of equivalent linear motion for the

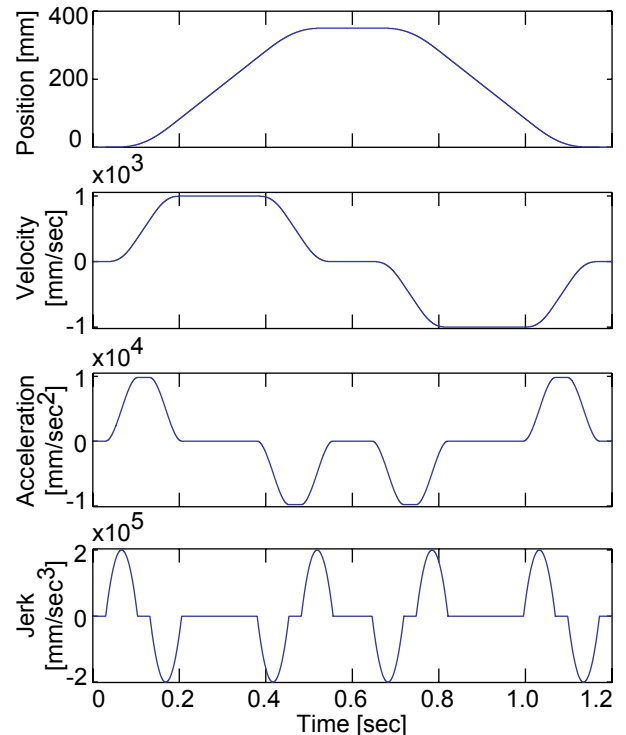


Figure 7: Command trajectory used in tracking tests.

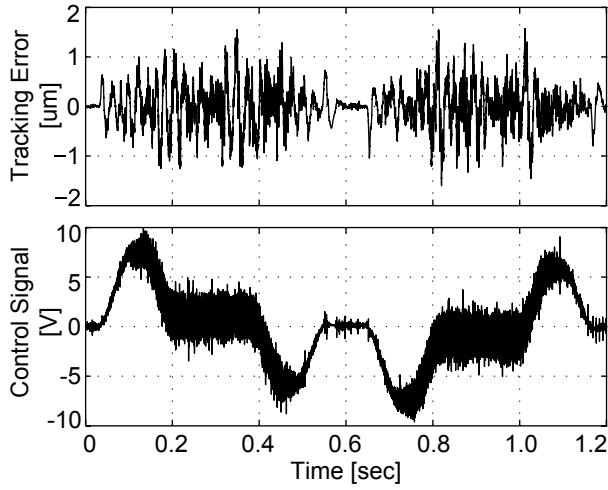


Figure 8: Experimental tracking performance.

commanded high speed trajectory.

In reality, the twist on the ball screw and the hysteresis type of motion loss in the preloaded nut [21] bring additional positioning error in the final motion delivered to the table. In the scope of this paper, the target has been to achieve high speed and high accuracy angular positioning of the ball screw. The mentioned factors that contribute to the overall positioning inaccuracy will be addressed in future work.

The reported servo performance could be achieved after proper design of notch filters, and implementation of active damping and feedforward friction compensation techniques. When the vibration attenuation components were not used, the sliding surface could only be increased up to 450 rad/sec (72 Hz), which resulted in a maximum tracking error of 11 μm for the same trajectory. For conciseness, this result is not presented here.

The contribution of active damping to motion smoothness is also demonstrated. Figure 9 shows the experimentally measured tracking transfer functions through Encoders 1 and 2. As mentioned earlier, Encoder 2 is the principal feedback in closing the position loop. The difference between Encoders 1 and 2 is used in generating the vibration cancellation signal. Considering Figure 9, a positioning bandwidth of 223 Hz has been achieved in the tracking response measured through Encoder 2. The application of active damping does not seem to alter the closed loop dynamics registered from this encoder significantly. Considering the frequency response measured through Encoder 1, the contribution of active damping in reducing the first torsional resonance at 445 Hz to half of its magnitude can clearly be seen.

This result was also verified by monitoring the position difference (i.e. twist) between Encoders 1 and 2 during the trajectory tracking experiments. Measurements of ball screw twist recorded without and with active damping are shown in Figure 10. As expected, the twist profile bears resemblance to the linear combination of the velocity and acceleration commands shown in Figure 7, indicating the presence of elastic deformation due to actuation torque as being the primary cause. Following elastic twist, the largest motion component is the vibration registered at the first torsional frequency (445 Hz), particularly when active damping is not used. With the application of active damping, these vibrations are significantly attenuated, thus resulting in smoother motion of the ball screw drive mechanism.

One particular advantage of the proposed control strategy is its robustness against changes in the drive's axial configuration. During frequency domain identification

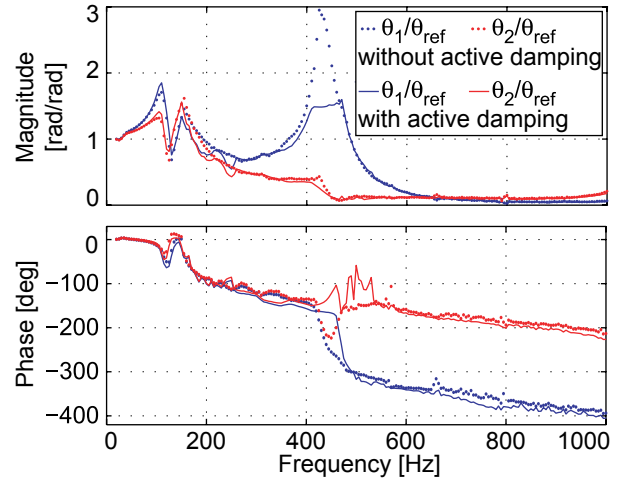


Figure 9: Closed loop tracking FRF's without and with active damping.

tests, it was seen that the natural frequencies and damping ratios of the torsional modes varied very little with changes in the axial position of the table and the excitation amplitude. This result was also confirmed in another study on a different feed drive system [25], indicating that axially transmitted reaction forces have only a minor influence on the torsional dynamics of ball screw drives. Hence, the closed loop response achieved by carefully modeling and cancelling out the torsional vibrations would not be significantly altered during the normal course of the drive's operation, particularly in the presence of axial position changes and external cutting and friction force disturbances.

5 CONCLUSIONS

This paper has presented a high bandwidth controller design for ball screw drives. The main control law is derived from adaptive sliding mode control, to achieve high robustness and tracking accuracy. Feedforward friction compensation is used to improve the accuracy at motion reversals. The closed loop bandwidth is widened using a combination of active and passive damping techniques to handle the first and second torsional modes of the ball screw. A servo accuracy of 1.6 μm has been achieved in tracking a trajectory with 1000 mm/sec feed and 1 g acceleration. Application of vibration attenuation has enabled a positioning bandwidth of 223 Hz to be

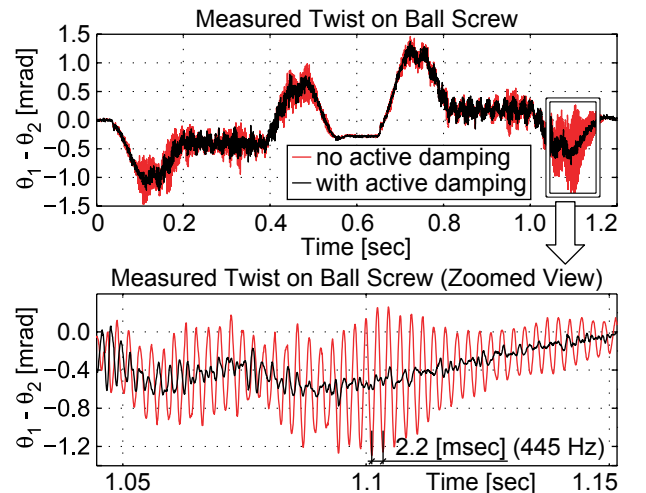


Figure 10: Contribution of active damping in attenuating ball screw torsional vibrations.

realized. Due to the complexity of the motion delivery between the ball screw and preloaded nut, the focus in this paper has been limited to improving the angular positioning of the ball screw. Improvement of linear positioning accuracy will be addressed in the continuation of this research.

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